1. Запуск landing_to_bronze.py

```
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - Обробка таблиці athlete_bio...
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - +----------+----------------+------+----------------+------+------+-----------------+-----------+--------------------+--------------------+
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |athlete_id|            name|   sex|            born|height|weight|          country|country_noc|         description|       special_notes|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - +----------+----------------+------+----------------+------+------+-----------------+-----------+--------------------+--------------------+
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |     65649|    Ivanka Bonova|Female|    4 April 1949|   166|    55|         Bulgaria|        BUL|Personal Best: 40...|                NULL|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |    112510|Nataliya Uryadova|Female|   15 March 1977|   184|    70|Russian Federation|        RUS|                NULL|Listed in Olympia...|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |    114973|Essa Ismail Rashed|  Male|14 December 1986|   165|    55|            Qatar|        QAT|Personal Best: 10...|Listed in Olympia...|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |     30359|      Péter Boros|  Male| 12 January 1908|  NULL|  NULL|          Hungary|        HUN|Between 1927 and ...|                NULL|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |     50557|   Rudolf Piowatý|  Male|   28 April 1900|  NULL|  NULL|    Czechoslovakia|        TCH|Rudolf Piowaty jo...|                NULL|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |    146111|Svetlana Kholomina|Female| 9 November 1997|  NULL|  NULL|              ROC|        ROC|                NULL|                NULL|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |    133041|  Vincent Riendeau|  Male|13 December 1996|   178|    68|           Canada|        CAN|                NULL|Listed in Olympia...|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |    110425|      Tanja Morel|Female|  4 October 1975|   164|    58|      Switzerland|        SUI|                NULL|Listed in Olympia...|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |    110705|   Maksim Shabalin|  Male| 25 January 1982|   183|    76|Russian Federation|        RUS|                NULL|Listed in Olympia...|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |     54541|        Gé Regter|  Male|   6 March 1916|  NULL|  NULL|      Netherlands|        NED|                NULL|                NULL|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - +----------+----------------+------+----------------+------+------+-----------------+-----------+--------------------+--------------------+
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - only showing top 10 rows
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO -
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - Таблицю athlete_bio успішно збережено в Bronze (Parquet) за шляхом: /home/marichka/datalake_project/data/bronze/athlete_bio
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - Завантаження https://ftp.goit.study/neoversity/athlete_event_results.csv...
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - Збережено в /home/marichka/datalake_project/data/landing/athlete_event_results.csv
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - Обробка таблиці athlete_event_results...
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - +--------------------+----------+-----------+--------+----------------+---------+----------------+----------+----------+-----+-----------+
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |            edition|edition_id|country_noc|   sport|           event|result_id|        athlete|athlete_id|       pos|medal|isTeamSport|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - +--------------------+----------+-----------+--------+----------------+---------+----------------+----------+----------+-----+-----------+
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|  100 metres, Men|    56265|Ernest Hutcheon|     64710|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|  400 metres, Men|    56313|   Henry Murray|     64756|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|  800 metres, Men|    56338|  Harvey Sutton|     64808| 3 h8 r1/2| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|  800 metres, Men|    56338|    Guy Haskins|    922519|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|  800 metres, Men|    56338|   Joseph Lynch|     64735|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|  800 metres, Men|    56338|   Henry Murray|     64756|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|1,500 metres, Men|    56349|   Joseph Lynch|     64735| 5 h2 r1/2| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|1,500 metres, Men|    56349|  Charles Swain|     79576|AC h3 r1/2| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|1,500 metres, Men|    56349|    Guy Haskins|    922519|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - |1908 Summer Olympics|         5|        ANZ|Athletics|1,500 metres, Men|    56349|   George Blake|     64619|       DNS| NULL|      false|
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - +--------------------+----------+-----------+--------+----------------+---------+----------------+----------+----------+-----+-----------+
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - only showing top 10 rows
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO -
[2026-04-25, 22:43:39 UTC] {subprocess.py:93} INFO - Таблицю athlete_event_results успішно збережено в Bronze (Parquet) за шляхом: /home/marichka/datalake_project/data/bronze/athlete_event_results
[2026-04-25, 22:43:39 UTC] {subprocess.py:97} INFO - Command exited with return code 0
```

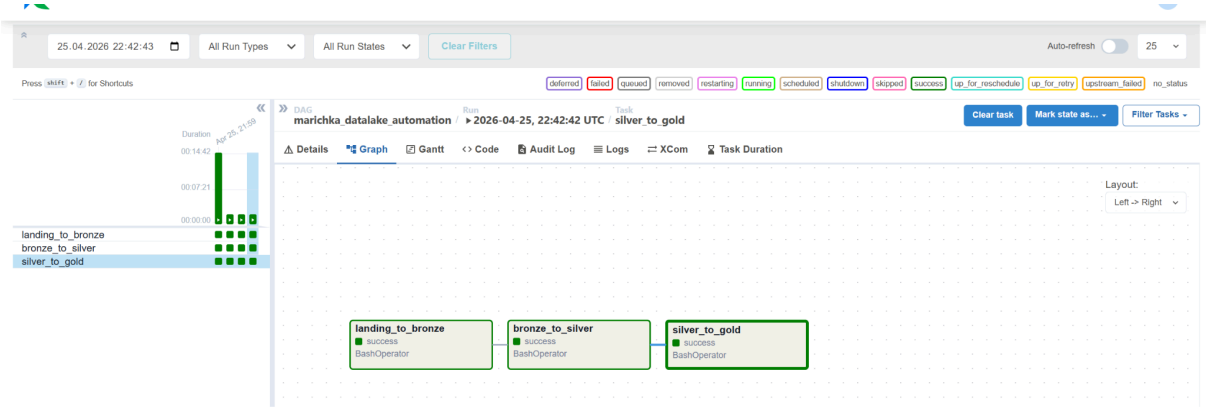
2. Запуск bronze_to_silver

```
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - Обробка athlete_bio: Bronze -> Silver...
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - +----------+----------------+------+----------------+------+------+-----------------+-----------+--------------------+--------------------+
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |athlete_id|            name|   sex|            born|height|weight|          country|country_noc|         description|       special_notes|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - +----------+----------------+------+----------------+------+------+-----------------+-----------+--------------------+--------------------+
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     72355|     Junko Asari|Female|22 September 1969|   164|    42|            Japan|        JPN|Personal Best: Ma...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     60628|       Andy Fitch|  Male|   28 August 1936|   163|    57|    United States|        USA|After prepping at...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     69306|  Adrian Metcalfe|  Male|    2 March 1942|   188|    87|    Great Britain|        GBR|Adrian Metcalfe w...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |    114938|  Mickaël Hanany|  Male|   25 March 1983|   198|    78|           France|        FRA|Personal Best: HJ...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     32945|     Dawn Marple|Female| 1 November 1970|   172|    64|    United States|        USA|Dawn Marple was a...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     20985|   Jana Angelakis|Female|  1 January 1962|   162|    63|    United States|        USA|Jana Angelakis wa...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     66740|  Surella Morales|Female|   16 June 1963|   167|    54|             Cuba|        CUB|Personal Best: 40...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     70866|Stavros Tziortzis|  Male|15 September 1948|   175|    64|           Greece|        GRE|Personal Best: 40...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     54062|     Lars Tomanek|  Male| 2 November 1965|   190|    95|          Germany|        GER|German water polo...|Listed in Olympia...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |     83787|   Igor Badamshin|  Male|    12 June 1966|   175|    70|Russian Federation|        RUS|Known mostly as a...|Husband of Nina G...|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - +----------+----------------+------+----------------+------+------+-----------------+-----------+--------------------+--------------------+
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - only showing top 10 rows
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO -
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - Таблицю athlete_bio збережено в Silver: /home/marichka/datalake_project/data/silver/athlete_bio
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - Обробка athlete_event_results: Bronze -> Silver...
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - +--------------------+----------+-----------+-------------------+-------------------+---------+-------------------+---------+---+-----+-----------+
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |            edition|edition_id|country_noc|             sport|              event|result_id|            athlete|athlete_id|pos| medal|isTeamSport|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - +--------------------+----------+-----------+-------------------+-------------------+---------+-------------------+---------+---+-----+-----------+
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1996 Summer Olympics|        24|        BLR|           Shooting|Free Pistol, 50 m...|    51881|      Ihar Basinski|    41345|  2|Silver|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1960 Summer Olympics|        15|        ETH|          Athletics|       Marathon, Men|    59715|        Abebe Bikila|    67392|  1| Gold|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1968 Summer Olympics|        17|        ETH|          Athletics|       Marathon, Men|    60599|         Mamo Wolde|    67434|  1| Gold|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1992 Summer Olympics|        23|        ETH|          Athletics|   5,000 metres, Men|    64041|         Fita Bayesa|    67385|  3|Bronze|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |2020 Summer Olympics|        61|        GEO|               Judo|   Lightweight, Men| 18001004|Lasha Shavdatuash...|   121786|  2|Silver|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1996 Summer Olympics|        24|        CUB|           Swimming|100 metres Backst...|     8023|       Neisser Bent|    46219|  3|Bronze|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1968 Summer Olympics|        17|        GDR|Artistic Gymnastics|Team All-Around, ...|    70322|      Erika Zuchold|    28524|  3|Bronze|       true|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1968 Summer Olympics|        17|        GDR|Artistic Gymnastics|Team All-Around, ...|    70322|         Karin Janz|    28502|  3|Bronze|       true|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1972 Summer Olympics|        18|        GDR|          Athletics|20 kilometres Rac...|    61144|      Peter Frenkel|    69782|  1| Gold|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - |1980 Summer Olympics|        20|        GDR|          Athletics|       High Jump, Men|    62299|      Jörg Freimuth|    69780|  3|Bronze|      false|
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - +--------------------+----------+-----------+-------------------+-------------------+---------+-------------------+---------+---+-----+-----------+
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO - only showing top 10 rows
[2026-04-25, 22:43:53 UTC] {subprocess.py:93} INFO -
```

3. Запуск silver_to_gold

```
Gold збережено: /home/marichka/datalake_project/data/gold/avg_stats
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - --- ТАБЛИЦЯ GOLD ---
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - +-------------------+------+------+-----------+-----------------+------------------+--------------------+
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |              sport| medal|   sex|country_noc|       avg_weight|        avg_height|           timestamp|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - +-------------------+------+------+-----------+-----------------+------------------+--------------------+
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |Equestrian Jumping|  Gold|  Male|        USA|71.666666666666667|182.11111111111111|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |        Ice Hockey|Bronze|  Male|        CZE|            89.84|            185.28|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |           Archery|Silver|  Male|        FRA|             83.0|             180.0|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |  Art Competitions|Bronze|  Male|        GER|             NULL|              NULL|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |   Figure Skating|Bronze|  Male|        HUN|             63.0|             168.0|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |          Shooting|Silver|  Male|        FRA|             75.8|             179.6|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |           Sailing|  NULL|  Male|        ITA|78.32432432432432|179.26351351351352|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |         Athletics|  NULL|  Male|        DOM|68.03174603174604| 174.0952380952381|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |              Judo|  NULL|  Male|        USA|84.79310344827586|179.43103448275863|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - |      Canoe Slalom|  NULL|Female|        CZE|             59.0|165.44444444444446|2026-04-25 22:44:...|
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - +-------------------+------+------+-----------+-----------------+------------------+--------------------+
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO - only showing top 10 rows
[2026-04-25, 22:44:05 UTC] {subprocess.py:93} INFO -
[2026-04-25, 22:44:05 UTC] {subprocess.py:97} INFO - Command exited with return code 0
```

4. Запуск графу project_solution через airflow


25.04.2026 22:42:43
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred
failed
queued
removed
restarting
running
scheduled
shutdown
skipped
success
up_for_reschedule
up_for_retry
upstream_failed
no_status
Duration
Apr 25, 21:59
00:14:42
00:07:21
00:00:00
landing_to_bronze
bronze_to_silver
silver_to_gold
DAG
marichka_datalake_automation
Run
2026-04-25, 22:42:42 UTC
Task
silver_to_gold
Clear task
Mark state as...
Filter Tasks
Details
Graph
Gantt
Code
Audit Log
Logs
XCom
Task Duration
Layout:
Left -> Right
landing_to_bronze
success
BashOperator
bronze_to_silver
success
BashOperator
silver_to_gold
success
BashOperator